MCO-1D Metachronal Cascade Ornithopter

ODrive Configuration Reference

GoolRC 5010 360KV · 7:1 Capstan Drive · 6S LiPo (22.2 V)

All values derived from first principles: KV, pole pairs, wing inertia, smootherstep acceleration profile at maximum frequency (3 Hz). Settings marked **red** are the primary cause of the 2 Hz frequency ceiling and must be changed immediately. Settings marked **green** are safe defaults that may be left as-is initially.

1. Motor Identity

These parameters tell the ODrive what motor it is driving. **pole_pairs** is the most critical — a wrong value produces correct-looking slow rotation but wrong commutation under load: motor will be weak, noisy, and very hot. Count rotor magnets physically and divide by 2. The GoolRC 5010 has 14 magnets = 7 pole pairs.

Setting	Value	Physical basis / why
`motor.config.motor_type`	**MOTOR_TYPE_HIGH_CURRENT**	Not GIMBAL — gimbal uses different current sensing, wrong for this motor
`motor.config.pole_pairs`	**7**	14 rotor magnets ÷ 2 = 7. Wrong value → no torque under load.
`motor.config.kv_rating`	**360**	Nameplate KV. KT = 60/(2π×KV) = 0.02653 Nm/A
`motor.config.calibration_current`	**5.0 A**	Used only during MOTOR_CALIBRATION state. Safe for 5010.
`motor.config.resistance_calib_max_voltage`	**4.0 V**	Max voltage applied during resistance measurement.
`motor.config.requested_current_range`	**30.0 A**	Configures ADC gain. Must be ≥ current_lim or sensing saturates.

2. Current Limits ← Fix for the 2 Hz Ceiling

The smootherstep deceleration at 3 Hz demands a peak current of **15.6 A** at the motor shaft (0.3957 Nm inertial + 0.0180 Nm aerodynamic, referred through 7:1). The ODrive default current_lim of ~10 A cannot supply this. The motor runs at its current ceiling on every deceleration — constant I^2R heating — and cannot accelerate fast enough to exceed 2 Hz. Setting current_lim to 25 A gives a 1.6× margin above peak demand and will allow 3 Hz immediately.

Setting	Value	Physical basis / why
`motor.config.current_lim`	**25.0 A**	Peak demand at 3 Hz = 15.6 A. 25 A = 1.6× margin. Motor rated ~28 A peak. ← THIS IS THE PRIMARY FIX.
`motor.config.current_lim_margin`	**5.0 A**	ODrive adds this before faulting → allows 30 A for brief reversal spikes.
`motor.config.current_control_bandwidth`	**2000 rad/s**	Sets inner PI loop speed. 2000 rad/s = 318 Hz >> 123 Hz electrical freq at peak. Kp = BW×L, Ki = BW×R (computed automatically after calibration).

3. Velocity Limits

Peak motor shaft velocity at 3 Hz (smootherstep): 110.0 rad/s = 1050 RPM (only 13% of no-load speed — back-EMF is only 2.92 V, 13% of supply. Motor is NOT voltage-limited).

Setting	Value	Physical basis / why

`controller.config.vel_limit`	**150 rad/s**	Peak demand = 110 rad/s. 150 = 1.4× margin. ODrive faults on overspeed if below actual peak. Default (~20–40) is almost certainly triggering faults now.
`controller.config.vel_limit_tolerance`	**16 rad/s**	15% of peak — allows coasting through reversal without fault.

4. Controller Gains (starting values — tune after calibration)

Bandwidth cascade: position BW → velocity BW → current BW, each 3× lower than the next inner loop. If oscillation: reduce vel_gain 30%. If sluggish: increase pos_gain 20%. Keep vel_integrator_gain low — integrator windup on a reversing drive causes instability.

Setting	Value	Physical basis / why
`controller.config.pos_gain`	**66.7**	Position BW = 67 rad/s. Units: (rad/s) / rad.
`controller.config.vel_gain`	**200.0**	Velocity BW = 200 rad/s. Units: A / (rad/s).
`controller.config.vel_integrator_gain`	**40.0**	Conservative — prevents windup on reversing drive. Units: A·s / rad.

5. Bus Voltage and Regenerative Braking

dc_max_negative_current is almost certainly not set correctly and is a major source of heat. At every stroke reversal the motor acts as a generator. If negative current is not allowed, the ODrive cannot sink the regenerative current and the bus voltage spikes — causing an overvoltage fault or forcing the energy into motor heat. Setting –25 A allows regen current to flow back to the battery.

Setting	Value	Physical basis / why
`config.dc_bus_overvoltage_trip_level`	**26.5 V**	6S max charge = 25.2 V. 26.5 V gives 1.3 V headroom for regen spike.
`config.dc_bus_undervoltage_trip_level`	**18.0 V**	3.0 V/cell × 6 = 18 V minimum before LiPo damage.
`config.dc_max_negative_current`	**–25.0 A**	Allows regen braking current back to battery. If bench supply (no regen): set to –0.01 and add braking resistor. ← CRITICAL
`config.enable_brake_resistor`	**False**	Set True only if using a braking resistor instead of battery regen.

6. Control Mode

For the MCO smootherstep profile: RPi Zero 2W sends position commands at 50 Hz. For rattan rod bench oscillation testing: velocity control with a sinusoidal input is simpler.

Setting	Value	Physical basis / why
`controller.config.control_mode`	**POSITION_CONTROL**	RPi sends position increments at 50 Hz. VELOCITY_CONTROL for bench sweep tests.
`controller.config.input_mode`	**INPUT_MODE_POS_FILTER**	Smooths 50 Hz position steps. Reduces jerk in current commands.
`controller.config.enable_vel_limit`	**True**	Keep enabled — protects motor from runaway.

`controller.config.enable_overspeed_error`	**True**	Keep enabled — faults on overspeed.

7. Watchdog

Setting	Value	Physical basis / why
`axis.config.watchdog_timeout`	**0.0**	Disabled for bench testing. Set to 1.0 s in flight — motor stops if RPi crashes.

8. Calibration Verification

After saving and rebooting, run calibration states in order. Check the measured values — if they fall outside the expected ranges, do not proceed to closed-loop operation.

Setting	Value	Physical basis / why
`motor.config.phase_resistance (after calib)`	**0.05–0.09 Ω**	Measured automatically. Above 0.15 Ω = wiring fault or wrong motor_type.
`motor.config.phase_inductance (after calib)`	**10–25 µH**	Measured automatically. Near-zero = calib_current too low (increase to 8 A).

9. Complete odrivetool Command Sequence

Paste the following into odrivetool in one block. Run once — settings persist after save_configuration().

```
# ■■ Motor identity ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
odrv0.axis0.motor.config.motor_type = MOTOR_TYPE_HIGH_CURRENT
odrv0.axis0.motor.config.pole_pairs = 7
odrv0.axis0.motor.config.kv_rating = 360
odrv0.axis0.motor.config.calibration_current = 5.0
odrv0.axis0.motor.config.resistance_calib_max_voltage = 4.0
odrv0.axis0.motor.config.requested_current_range = 30.0

# ■■ Current limits (primary fix for 2 Hz ceiling) ■■■■■■■■■■■■
odrv0.axis0.motor.config.current_lim = 25.0
odrv0.axis0.motor.config.current_lim_margin = 5.0
odrv0.axis0.motor.config.current_control_bandwidth = 2000

# ■■ Velocity limits ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
odrv0.axis0.controller.config.vel_limit = 150
odrv0.axis0.controller.config.vel_limit_tolerance = 16

# ■■ Controller gains (starting values — tune after calib) ■■■■
odrv0.axis0.controller.config.pos_gain = 66.7
odrv0.axis0.controller.config.vel_gain = 200.0
odrv0.axis0.controller.config.vel_integrator_gain = 40.0

# ■■ Bus voltage and regenerative braking ■■■■■■■■■■■■■■■■■■■■■■■
odrv0.config.dc_bus_overvoltage_trip_level = 26.5
odrv0.config.dc_bus_undervoltage_trip_level = 18.0
odrv0.config.dc_max_negative_current = -25.0

# ■■ Control mode ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
odrv0.axis0.controller.config.control_mode = CONTROL_MODE_POSITION_CONTROL
odrv0.axis0.controller.config.input_mode = INPUT_MODE_POS_FILTER
odrv0.axis0.controller.config.enable_vel_limit = True
odrv0.axis0.controller.config.enable_overspeed_error = True

# ■■ Watchdog (disabled for bench test) ■■■■■■■■■■■■■■■■■■■■■■■■
odrv0.axis0.config.watchdog_timeout = 0.0

# ■■ Save and reboot ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
odrv0.save_configuration()
odrv0.reboot()
```

```
# ■■ After reboot: calibration sequence ■■■■■■■■■■■■■■■■■■■■■■■■■
odrv0.axis0.requested_state = AXIS_STATE_MOTOR_CALIBRATION
# Wait ~5 s for two beeps then IDLE

odrv0.axis0.requested_state = AXIS_STATE_ENCODER_OFFSET_CALIBRATION
# Wait ~10 s

odrv0.axis0.requested_state = AXIS_STATE_CLOSED_LOOP_CONTROL

# ■■ Verify ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
print(odrv0.axis0.motor.config.phase_resistance) # expect 0.050–0.090 Ohm
print(odrv0.axis0.motor.config.phase_inductance) # expect 10e-6 – 25e-6 H
print(odrv0.axis0.motor.error) # expect 0 (no errors)
```

Thermal note: The GoolRC 5010 has no temperature sensor output. At 3 Hz sprint the $I_{rms} \approx 9.9$ A, generating ~5.8 W of heat per phase. Without airflow the stator will exceed 80°C in approximately 2–3 minutes. Add a 50 mm 5 V fan blowing axially across the stator for unlimited duration. Limit sprint bursts to 30 s per minute without airflow.

Bench supply warning: Most bench power supplies cannot sink regenerative current. If testing without a LiPo, set dc_max_negative_current = –0.01 and add a 2 Ω / 50 W braking resistor across the DC bus, or the bus voltage will spike at every reversal.